

Azlan Naeem

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Science in Computer Science

Sep 2022 – Apr 2027

- Enrolled in Co-op Program (ASIP/PEY)
- Schulich Leader Scholarship: Awarded \$80,000 out of 300,000 candidates

EXPERIENCE

Fullstack Software Engineer Co-op

May 2025 – Jun 2026

Flywheel Digital

Toronto, ON

- Architected a distributed workflow orchestration service to replace a legacy third-party platform, using two-level fanout and CAS-based locking to coordinate execution across 1,000+ advertiser accounts.
- Owned an end-to-end data integration feature bringing two new Amazon data sources into the Audience Builder, authoring the system design and building automatic geo- and audience-type-aware dataset selection logic across 16 new SQL query variants and 8 audience types.
- Built and integrated a cross-service pipeline surfacing ML-model-derived incrementality metrics across Python/Django, Java/Spring, and Redshift, including schema migrations and production reporting across two UIs.
- Built a unified Datadog dashboard tracking 20+ metrics across API endpoints, Airflow pipelines, and Cloud Tasks queues, adopted by 3+ teams for real-time debugging and incident response.

Machine Learning Student Researcher

May 2024 – Aug 2024

University of Toronto

Toronto, ON

- Built and trained CNN models in PyTorch to classify pediatric knee ultrasound images into age categories, developing an end-to-end preprocessing, training, and evaluation pipeline.
- Applied image preprocessing and augmentation techniques to address class imbalance and improve model generalization.
- Evaluated model performance using confusion matrices, ROC curves, and Grad-CAM to analyze prediction errors and interpret model decision regions.

PROJECTS

Food Classification ML Pipeline | Python, TensorFlow, scikit-learn, pandas

Feb 2025 – Apr 2025

- Developed an end-to-end ML pipeline comparing logistic regression, decision trees, neural networks, and Random Forest models; achieved 83.3% test accuracy.
- Engineered preprocessing for mixed-format survey data using one-hot/multi-hot encoding, ordinal mappings, and imputation; evaluated models using 5-fold stratified cross-validation, precision, recall, and F1-score.

Find My Keys | Python, Flask, React Native, SQLite, Docker, YOLO, AWS, Socket.io

Sep 2024 – Dec 2024

- Developed a full-stack personal item locator integrating YOLO object detection and voice transcription for image- and voice-based item retrieval.
- Implemented JWT authentication with refresh tokens and token blocklisting, and wrote pytest coverage for authentication, detection, and voice-command endpoints with GitHub Actions CI/CD.

Custom Threading Library | C, POSIX, x86-64

Oct 2024

- Built a preemptive user-level threading library in C with context switching, thread creation, scheduling, mutex locks, and synchronization using POSIX signals.
- Implemented FCFS and priority-based schedulers with blocking/wakeup mechanisms for concurrent execution.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, TypeScript, JavaScript, R

ML & Data: PyTorch, TensorFlow, scikit-learn, Keras, NumPy, pandas, Matplotlib

Frameworks & Databases: Django, Flask, Spring, React, React Native, PostgreSQL, SQLite, MongoDB, Redshift

Cloud & DevOps: AWS, Docker, Git, Datadog, Airflow, GitHub Actions